BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Smart Autonomous Vacuum & Wet Floor Mopping Robot

by

Group 01

Section 06

Group Members:

Name	ID
B M Rauf	22201782
Azmari Sultana	22201949
Sayeba Nasir	22201354

Department of Computer Science and Engineering

BRAC University

08/09/2025

Abstract

The project we proposed is a smart autonomous vacuum and wet floor mopping robot that is useful for minimizing workload in case of cleaning the house on a daily basis. The vacuum feature is useful for removing dust continuously and the mapping system will turn on when there is a wet surface detected by a moisture sensor with over 20% of moisture. To avoid obstacles like furniture, people and pets, ultrasonic sensors are used. Remote control and autonomous both modes are used to control the robot. Without compromising any kind of safety, remote control provides direct control for users. From enhancing hygiene to reducing dust and slipping risk because of any kind of spills, the system works. Specially, it helps the older people who have physical complexity to clean their house. It also provides safety measures to produce control systems and emergency stop button for allowing safe interaction with everyone. Overall, this robot provides a better home cleaning system which eases our daily cleaning routines

Keywords

Autonomous, Wet Floor Detection, Obstacle Avoidance, Embedded System, Sustainability

1. Introduction

Problem and Motivation: For elderly people, to keep their home’s floor clean is a physical task which can be difficult for them sometimes. They can face back pain or accidental slip anytime because of manual cleaning methods. This motivates the need for a semi automated solution that can reduce daily cleaning effort while improving hygiene and safety inside the home.

Objective: Design and build a smart autonomous vacuum and wet floor mopping robot that operates reliably in household environments.

Scope and Relevance to Computer Interface: It is relevant to computer interfacing as it integrates sensors, motor drivers and a microcontroller to coordinate multiple tasks. Interfacing the moisture sensor, ultrasonic sensors and Bluetooth module with the controller demonstrates how computer based systems can interact with the physical environment to create a practical, user friendly and safe cleaning device.

2. Related Work/ Inspiration

One of the robotic vacuums that significantly motivated research and development in the field of home cleaning automation is iRobot Roomba (iRobot, 2023). Although Roomba has advanced autonomous navigation and scheduled cleaning, it is usually costly and difficult. Our project aims at a simpler and less expensive home design. In contrast to Roomba, our robot is Bluetooth-controlled and it has basic autonomous control. It keeps the vacuum constantly active and performs mopping only when the floor is wet enough, which helps to save energy usage and prevent slips. Our project will make the system more accessible and affordable for ordinary households.

3. Technical Approach

System Architecture:

- **Navigation:** Autonomous and remote control using a Bluetooth module with ultrasonic sensors for obstacle avoidance and the necessary Arduino code.
- **Mopping Mechanism:** Servo motor controlled mopping system.
- **Vacuum Mechanism:** DC motor controlled vacuum system.

Components Used:

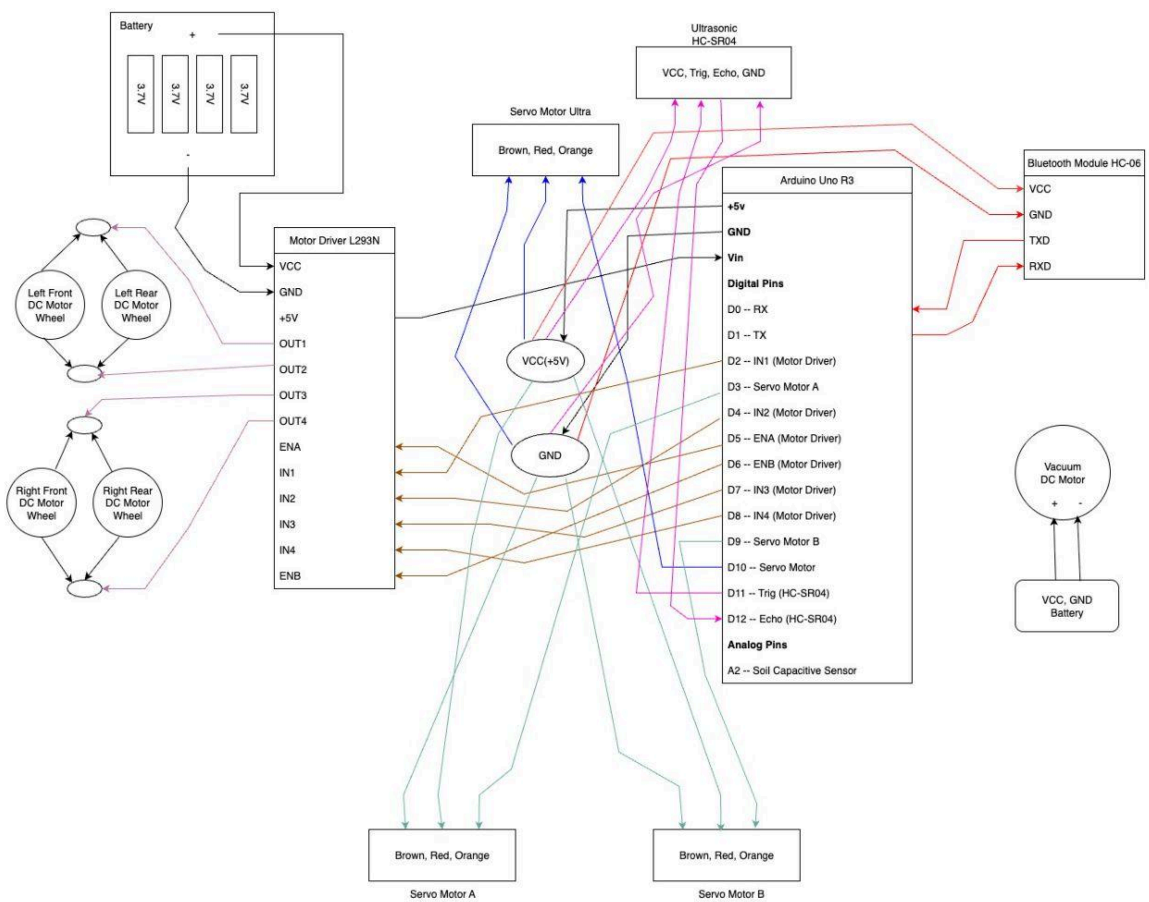
- **Hardware:** Arduino UNO R3, Ultrasonic sensor HC-SR04, Bluetooth Module HC-06, DC motor, Servo motor, Motor driver L298N, Capacitive soil moisture sensor
- **Software:** Arduino IDE, embedded C for microcontroller

Cost Breakdown:

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	Arduino Uno R3	1	988	988
2	DC Gear BO Motor	4	120	480
3	DC Motor with Fan	1	120	120
4	Capacitive Moisture Sensor	1	280	280
5	Motor Driver Module	1	168	168
6	Servo Motor	3	135	405
7	Ultrasonic Sensor HC-SR04	1	99	99

8	DC Motor Wheel	4	80	320
9	Bluetooth Module HC-06	1	330	330
9	Miscellaneous		1400	1500
Total Cost (BDT)				4690

Circuit/Schematic: **Forgot to add Soil Capacitive Sensor**



Functionality:

Our robot can be controlled in two ways:

1. Remote control mode (manual)
2. Autonomous mode (self-driving)

It can also detect floor moisture/wet and automatically start a mopping process when the ground is wet enough. The system combines motors, ultrasonic sensors, servos and a soil moisture sensor to complete its tasks.

Step by Step Explanation

System Start

- When the robot is powered on, it stays in pause mode.
- The user can send commands through the serial monitor or bluetooth module.

Modes of Operation

- Pause (P): The robot stops all actions.
- Start (A): User can now sends commands:
    - F: Move Forward
    - B: Move Backward
    - L: Turn Left
    - R: Turn Right

- Autonomous Mode (S): Robot drives on its own using ultrasonic sensors to avoid obstacles. If the user sends "S" again, it will get back to remote control mode.

Obstacle Detection

- The ultrasonic sensor scans the front, left and right.
- If an obstacle is closer than 40 cm (autonomous) or 25 cm (manual), the robot stops.
- In autonomous mode, it decides whether to turn left, turn right or go backward.

Wet Floor Check

- The soil sensor reads ground moisture/wet levels.
- If moisture/wet percentage is above 20%, the system starts the mopping process.

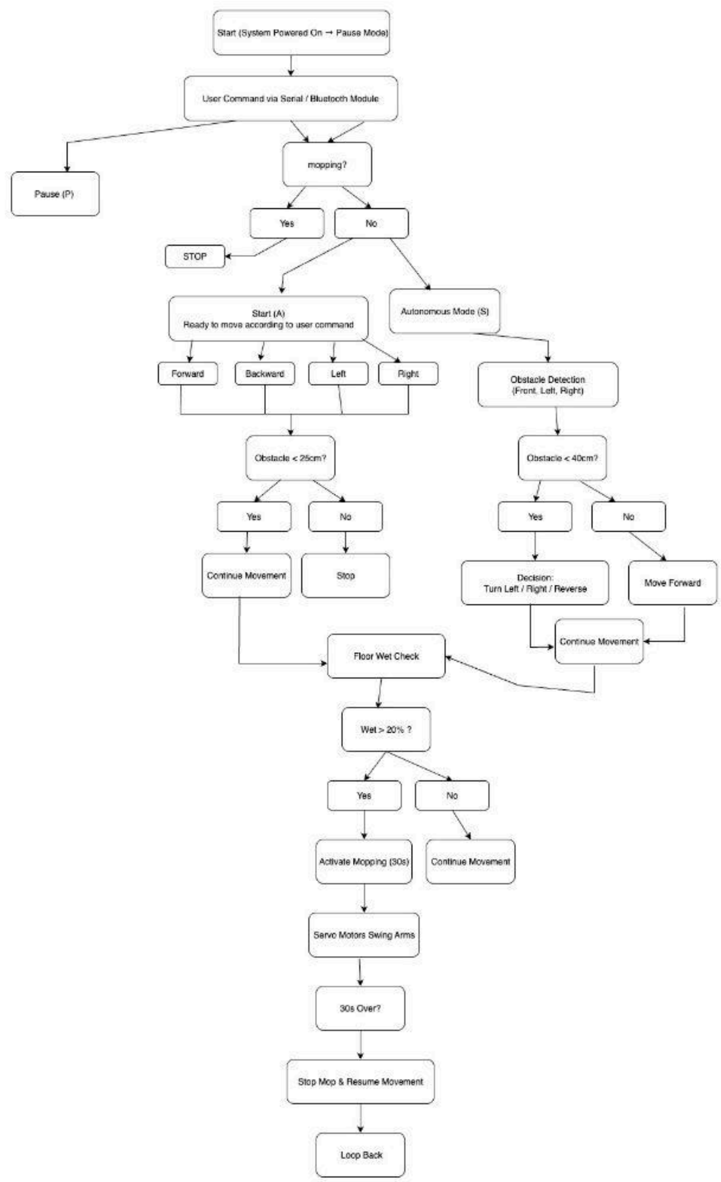
Mopping Mechanism

- Two servo motors move the mop arms back and forth between set angles (60deg-180deg).
- The mop runs for 30 seconds once activated, then stops automatically.

Motor Control

- Four DC motors (with L298N motor driver) control the wheels.
- Speed is set using PWM signals.
- Functions include moving forward, backward, left, right, swinging left/right (autonomous adjustment) and stopping.

Flow Chart



Challenges:

One major challenge was combining all the circuit components. The Arduino also did not have enough pins to support the sound sensor for sound-based control. To solve this, we added a Bluetooth module. With it, we can control the robot through a Bluetooth controller, which also removes the need for the sound sensor. Another challenge was a damaged DC motor, which we resolved by replacing it with a new one. The most important issue that we could not fully solve was in the autonomous mode. Since we used only one ultrasonic sensor (because of pin shortage on the Arduino), the robot could not detect obstacles properly from all sides. As a result, sometimes it collides with objects.

4. Sustainability & Impact

Sustainability: The robot will be energy-saving as it will only run when it is necessary, its mopping will only turn on when the floor is more than 20 percent wet and this will save energy. Rechargeable batteries reduce long term waste as opposed to disposable ones. The system is also eco-friendly because it can be constructed using lightweight and partly recyclable materials.

Impact: The robot can assist households in keeping their floors cleaner and safer with less effort, which is particularly beneficial to elderly users or individuals with mobility problems. It minimizes the repetition of work, saves time and avoids slip risks on wet floors. At the community level, it promotes the use of simple automation in everyday life and makes cleaning more participatory and more accessible.

Future Work: The robot can be enhanced with advanced autonomous navigation, mapping and zoning to clean more intelligently in the future. Scheduling and remote monitoring could be enabled through integration with mobile applications, smart home systems or IoT. More advanced sensors and more powerful motors will also enhance performance and reliability.

Limitations: Even though it was successfully implemented, the robot is not completely autonomous and does not support mapping or zoning. The other weakness is it couldn't detect objects from every side. Another limitation, the vacuum always remains on which consumes power.

5. Results & Discussion

The robot successfully performed the tasks it was designed for. During testing, it could move forward, backward, left, right and stop both manually and in autonomous mode. The ultrasonic sensor detected obstacles in front and to the sides, allowing the robot to avoid them by turning or reversing when needed. The soil moisture sensor also worked well and it activated the mopping mechanism when the floor was wet enough. The mopping arms moved smoothly in a back and forth motion using two servo motors and stopped automatically after 30 seconds which saved energy. The Bluetooth control allowed easy manual operation from a phone or bluetooth controller. However, since only one ultrasonic sensor was used, the robot sometimes failed to detect obstacles from all directions, leading to occasional bumps. Also, the vacuum ran continuously which consumed extra power. Despite these minor limitations, the robot's basic cleaning functions were effective and it worked reliably during testing, proving the idea is practical and usable for home environments.

6. Conclusion

In this project, a Smart Autonomous Vacuum and Wet Floor Mopping robot has been constructed to be used in the home to reduce the load of cleaning each day and enhance the level of cleanliness. There is continuous operation of vacuum and the mopping system would turn on when there is a wet floor over 20 percent. To ensure safety, it operates through bluetooth remote control and ultrasonic sensors. It provides safety by reducing slip risks and helps elderly people. The design is also environment friendly as it saves energy and can be useful to the house. Overall, it shows how low-cost simple robotics can simplify cleaning, make it safer and more sustainable.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
B M Rauf	22201782	Team Lead, Circuit Designing, Remote	- Circuit Implementation

		Controlled driving, Error debugging	- Write Code for Remote Control Driving - Code error debugging - Technical Support - Prepared components list and expenses
Azmari Sultana	22201949	Logic Implementation, Wet detection & Mopping, Flow Chart, Report Writing Support	- Wet Detection and Mopping logic and code implementation - Draw the flow chart of robot working procedure - Sustainability and Impact - Related Works and Inspiration
Sayeba Nasir	22201354	Hardware Designing, Autonomous Driving, Report Writing Support	- Connected Hardware - Autonomous Driving code implementation - Other parts of report writing

References

iRobot. (2023). *Roomba Robot Vacuum*. iRobot Corporation. https://www.irobot.com